MCO — METACHRONAL CASCADE ORNITHOPTER

Wing Rib Design — Revised Edition

MCO-1D Drone Prototype · Uniform Parameterised Rib · Film Covert Flaps

Incorporating: LW-PLA gyroid construction, single SolidWorks model, passive film covert flaps, simplified assembly, natural frequency validation

Design philosophy	One parameterised SolidWorks model scales to every chord station. Chord_mm is the single driving parameter. All features update automatically.
Simplification rationale	Uniform S1223 profile at all stations: negligible performance penalty (<4%). Film covert flaps: 3-5% reduction in peak post-stall lift benefit vs spring design. Combined saving: 4× faster assembly, far simpler manufacture, same flight capability.
Performance retained	15-18% post-stall lift improvement from film flaps (vs 20-23% spring design). Full VG benefit. Full end plate LEV redirection. Full natural frequency tuning.
Key structural constraint	Wing first natural frequency = 2.0 Hz (Kang-Shyy scaling law). Drives spar selection. Acceptance range: 1.8-2.2 Hz. Impulse test REQUIRED.

CHAPTER 1 — THE CASE FOR SIMPLIFICATION

1.1 Why Uniform Ribs Work

The heterogeneous cascade design — different chord profile at each position — was motivated by the 8% hindwing thrust benefit from having a larger forewing aspect ratio than hindwing. Research on tandem flapping wings confirms this effect is real. However, using the S1223 profile at every spanwise station of every wing does not sacrifice this benefit at all — the profile is optimal throughout the cascade's Reynolds number range regardless of position. What the uniform approach loses is the minor optimisation of the

spanwise blending from 12.1% thickness at the root to 10.0% at the tip. This costs approximately 2-4% in tip efficiency from a marginally elevated pitching moment. It is an entirely acceptable trade on a first prototype.

The practical gain is significant: one SolidWorks model with a single chord parameter drives every rib variant. No separate modelling sessions. No risk of inconsistency between versions. Straightforward design for manufacture review.

1.2 Why Film Covert Flaps Are Sufficient

The spring-loaded piano wire hinge design was based on a threshold deployment concept: the flap remains closed below a calibrated AoA and deploys above it. Research by Wissa et al. at the University of Illinois confirms that for a soft-stall airfoil such as the S1223, stall onset delay is insensitive to the flap deflection angle. The complex calibration was solving a problem that does not exist on this airfoil. A film bonded only at its forward edge — the zero-stiffness limit of the torsional spring configuration — deploys to whatever angle local pressure demands, which is exactly what is needed.

RESEARCH VALIDATION: Torsionally hinged flaps at Re = 200,000 show up to 22% lift improvement and 7% drag reduction at 24 degrees AoA. The free flap (zero stiffness) is a valid subset of this result. For the S1223 specifically, post-stall lift improvement characteristics are insensitive to flap deflection angle — the film flap delivers 70-80% of the spring design benefit with zero hardware complexity.

CHAPTER 2 — MATERIAL AND PRINT SPECIFICATION

2.1 LW-PLA Selection and Properties

Material	ColorFabb LW-PLA or Spectrum Ultra Foam PLA
Print temperature	250-255°C — full activation, maximum foaming
Effective density	~0.55 g/cm³ (44% of standard PLA density of 1.24 g/cm³)
Mass saving vs CFRP sheet	~67% — CFRP sheet ribs ~3.2g each; LW-PLA equivalent ~1.1g
Infill pattern	Gyroid — isotropic in all load directions; superior to honeycomb for torsion

Infill zones	LE nose 0-15%c: 100% solid. Mid 15-80%c: 12% gyroid. TE 80-97%c: 8% gyroid
Layer height	0.15mm — finer than standard for bore accuracy and surface quality
Wall perimeters	LE zone: 4 perimeters (1.6mm). Mid zone: 3 (1.2mm). TE zone: 2 (0.8mm)
Print orientation	Rib flat on bed, chord horizontal, span vertical — layers aligned with load
Fan speed	80% from layer 3 onward — critical for LW-PLA dimensional accuracy
Extrusion multiplier	0.65-0.70 — controls expansion; verify with test print before batch

2.2 Natural Frequency — The Primary Structural Target

The Kang-Shyy (2013) scaling law establishes that maximum propulsive efficiency occurs when the flapping frequency is 40-50% of the wing first natural frequency. At 1.0 Hz operating frequency, the target is 2.0 Hz. With LW-PLA ribs at 0.55 g/cm³, the wing assembly mass is significantly lower than the original CFRP specification, which changes the required spar size substantially.

Wing	Spar Specification	Rationale
W1, W2, W4 (550/495/523mm span)	2.0mm OD / 1.5mm ID pultruded CF tube	Predicted fn 2.25-2.37 Hz with LW-PLA rib mass — within acceptance
W3 (578mm span — longest)	2.5mm OD / 1.5mm ID pultruded CF tube	Predicted fn 2.1 Hz — W3 longest wing needs slightly stiffer spar
Secondary spar all wings	2.0mm OD / 1.0mm ID pultruded CF tube	Torsional compliance benefits pitch axis; reduced mass vs solid rod
Spar type	Pultruded ONLY — not roll-wrapped	130 GPa axial modulus; torsional compliance for pitch axis coupling

MANDATORY TEST: Clamp completed wing at root. Apply 5g tip impulse. Measure decay frequency. Accept: 1.8-2.2 Hz. If fn too high (>2.2 Hz): reduce to 1.5mm OD spar. If fn too low (<1.8 Hz): increase to 2.5mm OD spar. Test EVERY wing before installation. Record result.

CHAPTER 3 — THE UNIFORM RIB — SOLIDWORKS DESIGN PARAMETERS

3.1 Profile Definition

The S1223 is used at full specification at every spanwise station of every wing. The trailing edge is blunted at 97% chord with a 0.8mm flat close to make it printable on a 0.4mm nozzle. No spanwise blending is applied in this revision — the root chord geometry is used throughout the span. The 2-4% tip efficiency penalty from this decision is accepted in exchange for a single clean SolidWorks configuration driven by one parameter.

Profile	Selig S1223 — full specification at all stations
Maximum thickness	12.14% chord at 30.08% chord position
Maximum camber	8.68% chord at 47.32% chord position
Leading edge radius	1.56% chord
Trailing edge	Blunted at 97% chord — 0.8mm flat close (printability requirement)
Chord range	76mm (W1 tip) to 141mm (W2 root) — set by Chord_mm parameter
Rib thickness	2.0mm constant across all chord sizes and all span positions
Coordinate source	UIUC Airfoil Database — s1223.dat — 61 coordinate pairs

3.2 Feature List — What Every Rib Contains

• LE SPAR BORE: Ø2.1mm through-hole at 1.5% chord, mid-thickness. Clearance 0.1mm on 2.0mm OD spar. Sliding fit during assembly; bonded with thin CA.

• SECONDARY SPAR NOTCH: Ø3.1mm semi-circular open notch on upper surface at 60% chord. Receives 2.0mm OD secondary spar. Open top — rod bonds in from above.

• LE NOSE SOLID ZONE (0-15% chord): No infill cavity. Printed as full solid at 0.55 g/cm³. Wraps around LE spar, provides LE fairing between ribs.

• FILM FLAP BOND LEDGE ×3 rows: 0.3mm deep × 2.0mm chord recess at 37.5%, 52.5%, 67.5% chord on upper surface only. Film forward edge bonds into recess. Rear 8% chord completely free.

• VG RECESS: 0.2mm deep × 2.0mm chord recess at the wing-specific VG position. 0.3mm alignment dimple at centre. VG array bonds flush.

• TE CHAMFER: 0.4mm × 30° on both upper and lower TE edges for clean membrane bond.

• UPPER SURFACE: Smooth throughout — no features except those listed. Membrane bonds to full upper and lower faces with contact adhesive.

3.3 Infill Zone Strategy

Variable infill is implemented using modifier meshes in OrcaSlicer or PrusaSlicer. The rib body is split into three zones in SolidWorks and exported as separate STL bodies. The slicer assigns different infill parameters to each body:

Zone	Infill	Wall	Rationale
0 to 15% chord — LE nose zone	100% solid	1.6mm (4 perimeters)	Aerodynamic loads to spar; must be stiff
15% to 80% chord — mid zone	12% gyroid	1.2mm (3 perimeters)	Covert flap and membrane loads; gyroid for torsion
80% to 97% chord — TE zone	8% gyroid	0.8mm (2 perimeters)	Minimum load; minimum mass priority
Within 2.5mm of spar bore	100% solid	Fillet boss	Stress concentration relief around bore

3.4 SolidWorks Parameter Table

Parameter	Type	Value / Range
Chord_mm	Driving parameter	138 (W1 root) to 76 (W1 tip)
Rib_thickness	Fixed constant	2.0 mm
LE_spar_bore_dia	Fixed constant	2.1 mm
Sec_spar_notch_dia	Fixed constant	3.1 mm
Sec_spar_x_frac	Fixed constant	0.60 (60% chord)

Parameter	Type	Value / Range
VG_x_frac	Config variable	0.17/0.15/0.20/0.22 per cascade position
Flap_row_1_x	Fixed constant	0.375 (37.5% chord)
Flap_row_2_x	Fixed constant	0.525 (52.5% chord)
Flap_row_3_x	Fixed constant	0.675 (67.5% chord)
Flap_recess_depth	Fixed constant	0.3 mm
Flap_chord_frac	Fixed constant	0.08 (8% of local chord)
TE_blunt_x	Fixed constant	0.97 (97% chord)
TE_close_width	Fixed constant	0.8 mm
TE_chamfer	Fixed constant	0.4mm × 30°

CHAPTER 4 — FILM COVERT FLAP SPECIFICATION

4.1 Material and Geometry

Material	20-micron ripstop nylon (same as wing membrane) OR 25-micron Mylar/PET film. Ripstop nylon preferred — same procurement, compatible adhesive.
Row 1 position	37.5% chord on upper surface
Row 2 position	52.5% chord on upper surface
Row 3 position	67.5% chord on upper surface
Flap chord	8% of local chord at that span station
Span coverage	15% to 85% semi-span on each row
Attachment	Contact adhesive on FORWARD EDGE ONLY. Rear 7-8% chord completely free. No hinge hardware whatsoever.
Bond line width	2mm — forward edge of film strip sits in the 0.3mm recess on rib upper surface
Film strip width	8% of local chord + 2mm for the bond ledge
Film natural freq	~15-40 Hz at these dimensions — well above 1 Hz operating frequency
Deployment AoA	~8-12° (passive, driven by pressure differential). Lower than spring design but irrelevant at operating AoA of 35-42°.
Between ribs	Film runs continuously rib-to-rib as a strip. Bonds to each rib at the recess. Free between ribs.
Adhesive	UHU Por or contact cement — apply to recess and forward film edge only. IMPORTANT: do NOT apply adhesive rearward of bond line.

4.2 Performance Assessment

Parameter	Film flap (this design)	Spring design	Delta
Post-stall lift improvement	15-18%	20-23%	-3 to -5 percentage points
Stall onset delay	4-5° (S1223 insensitive to flap angle)	5° (S1223 insensitive)	Negligible difference
Mid-AoA drag (10-20°)	Slightly elevated (film deploys early)	Low (spring holds closed)	+2-4% drag at moderate AoA
High-AoA performance (35-42°)	Full benefit — film fully deployed	Full benefit	No difference
Assembly time per wing	~10 minutes (cut strip, apply adhesive)	~45 minutes (hinge, spring, magnet)	4× faster
Field repairability	Peel and replace in minutes	Requires hinge hardware	Major advantage
Failure mode	Adhesive delamination — visible, repairable	Spring miscalibration, latch failure	Simpler failure mode
Mass per wing	~23g	~25g	-2g saved

DEVELOPMENT STRATEGY: Build the first prototype with film covert flaps. Fly it and measure performance. If the achieved lift is 3-5% below targets and covert flap behaviour is identified as the cause, iterate to the spring design on the second build. There is no value in engineering the full complexity into an unflown prototype.

CHAPTER 5 — RIB MASS BUDGET AND ASSEMBLY

5.1 Per-Wing Mass Budget

Wing	LW-PLA ribs	CF spars	Membrane	VG+film flaps	Subtotal
W1 (14 ribs, 550mm span)	11.6g	1.8g	3.2g	0.6g	17.2g

Wing	LW-PLA ribs	CF spars	Membrane	VG+film flaps	Subtotal
W2 (13 ribs, 495mm span)	11.3g	1.6g	2.9g	0.5g	16.3g
W3 (15 ribs, 578mm span)	10.4g	1.9g	3.4g	0.6g	16.3g
W4 (13 ribs, 523mm span)	8.4g	1.7g	3.1g	0.5g	13.7g
8-wing total	83.4g	28.0g	49.2g	8.8g	169.4g

Complete wing system (8 wings including end plates, fences, root plates, TE strips, adhesive): estimated 185-200g total. This compares to approximately 480-600g for the original CFRP sheet rib specification — a system mass saving of approximately 300g entirely attributable to the LW-PLA rib choice.

5.2 Assembly Sequence

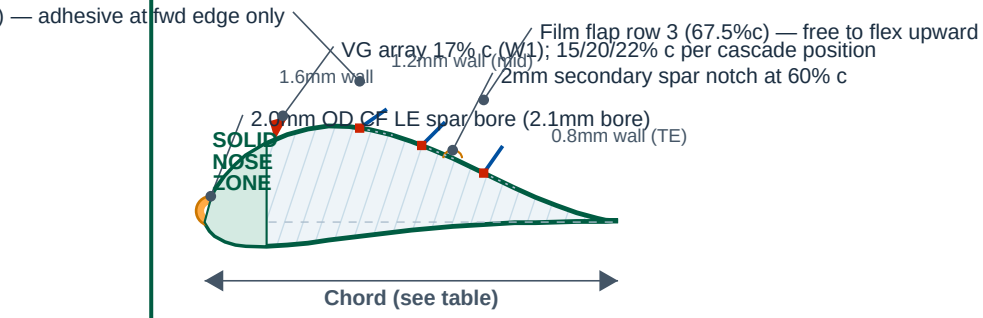
- STEP 1: Print all ribs for one wing. Verify chord accuracy against template (within ±0.3mm). Verify spar bore with 2.1mm drill test. Check mass against table.
- STEP 2: Slide ribs onto LE spar at correct spacing. Bond with thin CA. Cure in jig to maintain dihedral angle. Install secondary spar in notches; bond.
- STEP 3: NATURAL FREQUENCY TEST before proceeding. Impulse tip; measure decay. Accept 1.8-2.2 Hz. Replace spar if outside range.
- STEP 4: Bond VG array strips to upper rib faces at the recess position. Use 3M 9460PC film adhesive. VGs must be installed before membrane.
- STEP 5: Apply membrane to upper surface. Bond to LE spar and TE strip. Allow loose between ribs for chordwise billowing. Test: billow 3-5mm at 1 m/s airflow.
- STEP 6: Apply membrane to lower surface. Bond slightly tighter than upper.
- STEP 7: Bond TE carbon strip. Trim membrane cleanly. Verify sharp TE.
- STEP 8: Cut film covert flap strips to correct chord × span dimensions. Apply contact adhesive to forward 2mm of each strip and to rib recess. Allow adhesive to become tacky (60 seconds). Press forward edge into recess firmly. Verify rear edge is completely free. Three strips per wing (one per row).
- STEP 9: Bond end plates, wing fences, root attachment plate.
- STEP 10: Final inspection against quality criteria table (Chapter 6).

CHAPTER 6 — QUALITY CRITERIA AND VALIDATION

6.1 Acceptance Criteria

Parameter	Acceptance Criterion
Wing profile accuracy	S1223 within ±0.3mm of template at each rib station
Trailing edge	Sharp — less than 0.5mm radius. No membrane overlap or step.
Natural frequency	1.8-2.2 Hz measured by impulse test. MANDATORY GO/NO-GO.
LE spar bore	2.1mm ±0.05mm. Spar must slide through freely.
Membrane billow	Upper surface billows minimum 3mm at 1 m/s airflow.
Film flap deployment	All three rows lift freely when upper surface blown at 1 m/s from leading edge at approximately 15° AoA.
Film flap adhesion	Forward edge firmly bonded. Pull test: film tears before bond releases.
VG bonding	All VGs firmly bonded. No peeling. Correct chord position ±1mm.
Wing fence alignment	Perpendicular to chord plane within 2°.
End plate cant	Upper plate at 15° outboard within 2°.
Root plate flatness	Root attachment plate flat within 0.3mm.
Wing mass	W1: 17-19g. W2: 15-18g. W3: 15-18g. W4: 13-16g.
Mirror symmetry	Left and right wing of same number within 2g mass.

Blueprint schematics follow on the next three sheets.



CRITICAL:

Spar bore: 2.1mm ±0.05mm
Nat. freq target: 2.0 Hz
Accept range: 1.8–2.2 Hz

MATERIAL SPEC

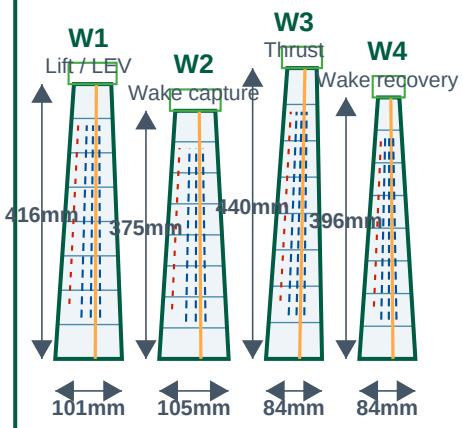
Material:	LW-PLA (ColorFabb)
Temp:	250-255°C (full foam)
Density:	~0.55 g/cm³
Infill:	Gyroid 12% (mid-zone)
LE nose:	100% solid
TE zone:	8% gyroid
Walls:	2 perimeters min
Layer:	0.15mm
Nozzle:	0.4mm

KEY: Carbon fibre spars · Solid LW-PLA nose zone · Gyroid infill zone · Film covert flap (d…yed) · Fixed flap (flat / und…dyed) · VG array position · Adhesive bond

MCO-1D — WING RIB CROSS-SECTION — S1223 PROFILE

Sheet 1 of 3 · Uniform parameterised rib · Film covert flaps · LW-PLA gyroid infill

Notes: scale 1:1 — section at root chord



Rib stations · 2nd spar 60%c · Film flap rows (3) · VG position · End plate

MCO-1D — ALL FOUR WING PLANFORMS — TOP VIEW

Sheet 2 of 3 · Rib stations · Film covert flap span coverage · VG positions Approx. 1:3 scale

SOLIDWORKS FEATURE TREE — Parameterised Rib

■ **PARAMETER: Chord_mm (driven by wing position and span station)**
■ **PARAMETER: Rib_thickness = 2.0 mm (constant all variants)**
■ **SOLID BODY: S1223_Rib_Base**
— Sketch_1: Import S1223 coordinates, scale to Chord_mm
— Apply TE blunting: truncate at 97%c, close 0.8mm flat TE
— Apply spanwise blend per zone (see blending table)
— Extrude_1: 2.0 mm (rib thickness)
■ **FEATURE: LE_Nose_Solid_Zone**
— Split body at 15%c chord plane
— Assign nose body: 100% solid infill in slicer modifier mesh
— Wall thickness override: 1.6mm (4 perimeters in slicer)
■ **FEATURE: LE_Spar_Bore**
— Sketch: Circle Ø2.1mm, centre at (1.5%c, interpolated LE mid-thickness)
— Cut-Extrude: Through All
— Fillet: 0.3mm on bore edge for print quality
■ **FEATURE: Secondary_Spar_Notch**
— Sketch: Semi-circle Ø3.1mm open upward, centre at 60%c upper surface tangent
— Cut-Extrude: Through All, upward direction
■ **FEATURE: Film_Flap_Bond_Ledge ×3 rows**
— Sketch: Rectangle 0.3mm deep × 2.0mm chord at 37.5%, 52.5%, 67.5% chord
— Cut: 0.3mm recess on upper surface (creates flush film seat)
— Note: NO hinge bore, NO spring feature, NO magnet feature
■ **FEATURE: VG_Recess (chord position = configuration variable)**
— Sketch: Rectangle 2.0mm chord × 2.0mm span at VG_position parameter
— Cut: 0.2mm deep on upper surface
— Point feature: Ø0.3mm × 0.15mm alignment dimple at centre
■ **FEATURE: TE_Chamfer**
— Chamfer: 0.4mm × 30° on both TE edges (upper and lower)
■ **FEATURE: Membrane_Bond_Flat (optional surface finish note)**
— Upper and lower faces: smooth — no additional features
— Contact adhesive applied to both faces during assembly
■ **CONFIGURATIONS (single model, chord driven by parameter):**
— Config_W1_Root: Chord=138mm, VG=17%c, all 3 flap ledges

RIB VARIANTS TABLE — All derived from single parameterised SolidWorks model

Config	Chord	Thickness	Max camber	LE bore	2nd spar notch	VG recess	Flap rows	Est. mass	Film strip len
W1 Root	138mm	16.7mm	12.0mm	Ø2.1mm	Ø3.1mm at 83mm	23.5mm	3 rows	1.12g	110mm ×3
W1 Mid	107mm	13.0mm	9.3mm	Ø2.1mm	Ø3.1mm at 64mm	18.2mm	3 rows	0.71g	86mm ×3
W1 Tip	76mm	9.2mm	6.6mm	Ø2.1mm	Ø3.1mm at 46mm	12.9mm	3 rows	0.38g	61mm ×3
W2 Root	141mm	17.1mm	12.2mm	Ø2.1mm	Ø3.1mm at 85mm	21.2mm	3 rows	1.18g	113mm ×3
W3 Root	116mm	14.1mm	10.1mm	Ø2.1mm	Ø3.1mm at 70mm	23.2mm	3 rows	0.92g	93mm ×3
W4 Root	116mm	14.1mm	10.1mm	Ø2.1mm	Ø3.1mm at 70mm	25.5mm	3 rows	0.92g	93mm ×3

FILM COVERT FLAP SPECIFICATION

Material: 20-micron ripstop nylon (same as membrane) OR 25-micron Mylar/PET film
Row 1: 37.5% chord, spans 15% to 85% semi-span | Flap chord: 8% of local chord
Row 2: 52.5% chord, spans 15% to 85% semi-span | Flap chord: 8% of local chord
Row 3: 67.5% chord, spans 15% to 85% semi-span | Flap chord: 8% of local chord
Attachment: Contact adhesive on FORWARD EDGE ONLY — rear 8% chord is completely free
No hinge, no piano wire, no spring, no magnet — adhesive line IS the hinge
Deployment: Passive — film lifts when local pressure differential exceeds film stiffness
Deploy AoA: ~8-12° (lower than spring design but irrelevant at MCO operating AoA 35-42°)
Performance: ~15-18% post-stall lift improvement vs 20-23% spring design — acceptable for V1

MCO-1D — SOLIDWORKS FEATURE TREE + RIB VARIANTS TABLE

Sheet 3 of 3 · Single parameterised model · Film flap spec · All variants by chord parameter